

Sriraj Paruchuru

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PROFILE

Software engineer and applied AI builder pursuing an **MSc in Artificial Intelligence & Machine Learning** at the University of Birmingham. Builds RAG applications and evaluates LLMs. Previously built pharmaceutical analytics software at Norstella for **nearly 18 months**. Seeking applied AI, software and forward deployed engineering roles; open to relocation.

TECHNICAL SKILLS

Programming & Applications: Python, TypeScript, JavaScript, React.js, Node.js, FastAPI

Applied AI: LLM APIs, retrieval-augmented generation (RAG), structured outputs

Development & Delivery: Docker, Azure DevOps, Git, Playwright

RELEVANT EXPERIENCE

Software Developer - Norstella

Apr 2024 - Sep 2025

Strategic Intelligence & Enterprise Value Platform teams | Pharmaceutical analytics SaaS

- Delivered **React, TypeScript and Node.js** features across **2 pharmaceutical analytics platforms**, taking assigned work from requirements through testing and **3-4-day release cycles**.
- Optimised **backend routing and server-side filtering** for queries scanning **millions of rows across multiple databases**, reducing filtered-data response latency from approximately **300-500 ms to 20-30 ms**.
- Built **Excel export, AWS Lambda and S3 data-delivery workflows** with validation and error handling; implemented **Auth0 SSO, JWT authentication and role-based access controls**.
- Contributed to the **design and backend development of internal AI tools alongside senior colleagues**, taking on responsibilities beyond assigned product work.

Student Activator - University of Birmingham

Apr 2026 - Jul 2026

Active Residences programme | Part-time, fixed-term contract

- Facilitated weekly sport and fitness sessions for **15+ students per session**, helping participants feel welcome and supporting inclusive participation across multiple sports.
- Took on additional responsibilities to **deliver Social Sport activities beyond the Active Residences programme**, collaborating with University of Birmingham Sport & Fitness teams.

SELECTED PROJECTS

More projects, with live demos: sriraj-p.github.io

One Size Might Not Fit All - MSc Dissertation (submitted Sep 2026)

Tools: Python, BM25, FastEmbed/ONNX, NumPy, RAG, pytest

- Designed and implemented a benchmark to test whether **LLM reliability varies by domain**, comparing **6 economical models across 350 cases in 5 domains**, including grant eligibility and contract interpretation.
- Built **keyword and semantic retrieval** to surface rules, exceptions and conflicting guidance from **34 public documents and 510 contracts**; added source-passage checks to validate generated citations.
- Executed **3,120 benchmark requests**, including repeat runs on a fixed 50-case subset and controls with evidence withheld; recorded model behaviour, estimated cost, latency and failures.
- Implemented uncertainty-based **answer, clarification and escalation paths**, plus a resumable runner with fixed prompts and saved configurations for reproducible model comparisons.

CrackD - AI Interview Preparation

Tools: Python, FastAPI, React, TypeScript, LLM APIs, RAG, structured outputs

- Developed a hackathon prototype into a **resume-analysis and mock-interview platform**, building backend and frontend workflows that compare CVs with job descriptions and tailor interviews to competency gaps.
- Used feedback from **25+ students and professionals** to refine **UX and interview scoring**; over **60% of testers** reported current use and intent to continue once fully built.
- Replaced regex score extraction with **strict JSON-schema outputs for 8 scoring fields and 4 feedback cards**; separately resolved report truncation and duplicate fallback displays.
- Turned feature ideas and tester feedback into scoped implementation and validation tasks within a **61-issue backlog**.

Retail Behavioural Anomaly Detection

Tools: Python, pandas, SciPy, statsmodels, scikit-learn

- Built and evaluated an **anomaly-detection pipeline** on **3,000 synthetic customers and 78,000 transactions** over 26 weeks, combining time-series decomposition and external signals to distinguish **9 behavioural categories**.

HACKATHONS & AWARDS

- **Champion & Most Valuable Product** - Boffin's Den, **Google Cloud Agentic AI Hackathon** (Mar 2026), for CrackD.
- **Placed 5th** - **Google Cloud Agentic AI Hackathon qualifier**: multi-agent CRM using ADK, Vertex AI, BigQuery and RAG.
- **Placed 4th** - **Google AI Playground Hackathon** (Google London, Dec 2025): multimodal Gemini app with team BrumMind.

EDUCATION & PROFESSIONAL DEVELOPMENT

Master of Science (MSc) in Artificial Intelligence & Machine Learning

Sep 2025 - Sep 2026
(expected)

University of Birmingham, UK

- LangChain, LangGraph & LangSmith Workshop - PwC x University of Birmingham (Feb 2026)

Bachelor of Technology (B. Tech) in Computer Science & Engineering

Jul 2019 - Sep 2023

Vellore Institute of Technology, AP, India